handle the incoming call, do some post-processing and sending the response back, you risk having a lot of 503[9] errors – and, guess what? That also goes against your SLA.

Maybe at this point is worth nothing that, often, if you are a paid service and you violate your SLA, you might have to give some money back to your customers. So, you definitely would like to avoid that, understandably.

Let's say, in order to keep things simple for our example, that you'd like to provide this sentiment analysis for one company at the time (one company per HTTP call), per day, and always only for the day before: imagine that people will use the information gathered yesterday, before the stock market opens, to decide whether or not it's a good idea to invest in a company.

Also, let's assume we only provide our prediction service for the Fortune top 100 companies, and just those.

7.2 First Attempt: Remembering Values

At this point, we know that we can't afford to compute each intermediate result over and again every time we receive a call, that would be too expensive. So, the most natural thing to do is storing these intermediate results once we compute them and looking them up when we need the results again.

We are obviously in luck because, since the function to produce the summary takes only one argument, the name